Category-Aware Explainable Machine Learning for Urban Service Demand Forecasting in Smart Cities

Tran Dai Phong Lam[1*], Thu Le[1], Nguyen Quoc Hung[1], Nguyen Trung Trinh[1]

[1*]FPT University, Ho Chi Minh City, Vietnam.

*Corresponding author(s). E-mail(s): phonglam2599@gmail.com;
Contributing authors: thulvm@fpt.edu.vn; hungtvt2222@gmail.com; trinhnguyen112355@gmail.com;

Abstract

Urban 311 systems record large-scale traces of reported municipal demand, but forecasting operationally useful surges remains difficult because reports are spatially uneven, service categories differ, and reporting behavior shapes observed counts. We formulate next-week abnormal reported-demand forecasting as a neighborhood-week-service-category early-warning task. Using New York City as a case study, the prospective model uses 30.9 million NTA-matched 311 requests with shifted historical demand, calendar timing, feature-week weather, service category, and borough identifiers. A target-week chronological protocol trains on 2015-2022 target weeks, validates on 2023, and tests on 2024-2025. The validation-selected decision strategy averages LightGBM and XGBoost scores and applies complaint-category-specific thresholds selected only on validation. On the held-out future test period, it obtains F1 = 0.3802, precision = 0.2933, recall = 0.5404, PR-AUC = 0.3310, and ROC-AUC = 0.7643. SHAP analysis of the prospective LightGBM component shows that historical temporal dynamics dominate, followed by service category, calendar, and weather features. Contemporary OSM and PLUTO snapshots are evaluated separately as retrospective context and are excluded from the prospective alerting model.

Keywords: urban computing, smart cities, 311 service requests, service-demand forecasting, explainable machine learning, SHAP, spatiotemporal modeling

1 Introduction

Urban service systems increasingly rely on administrative and citizen-generated data to monitor conditions, allocate attention, and anticipate operational pressure. NYC 311 is especially valuable because it records requests across infrastructure, housing, sanitation, public safety, environmental, and quality-of-life domains. Yet 311 counts are not direct measurements of all underlying urban problems. They reflect both conditions and the propensity to report them, which varies with access, awareness, trust, service expectations, and governance processes [1–4]. Forecasts built from these data therefore need careful wording: the target is abnormal reported service demand, not the true incidence of every urban issue.

We study a prospective early-warning task at the neighborhood tabulation area (NTA), week, and complaint-category level. For each feature week, the model estimates whether the following week will show abnormal reported demand relative to recent local history. The headline alerting model uses only predictors available by the end of feature week t: historical 311 demand, calendar timing, feature-week weather exposure, service category, and borough identifiers. Lags and rolling statistics are shifted so no target-week counts enter the feature matrix. Contemporary OSM and PLUTO snapshots are excluded from the final ensemble, thresholds, and headline metrics, and are assessed only in a separate retrospective context analysis.

The analysis examines whether next-week abnormal reported demand can be forecast under a strictly chronological future test, how service-specific thresholds alter precision, recall, and alert volume across complaint domains, and which forecast-available features contribute most strongly to model predictions. Its contributions are a leakage-controlled next-week forecasting task, a prospective feature set using information available at week t, target-week chronological evaluation, validation-only service thresholds, explainability of prospective model behavior, and a separate retrospective assessment of contemporary urban context. The practical aim is to prioritize analyst attention toward elevated reported-demand risk without treating 311 data as a complete map of urban need. Figure 1 summarizes the workflow.

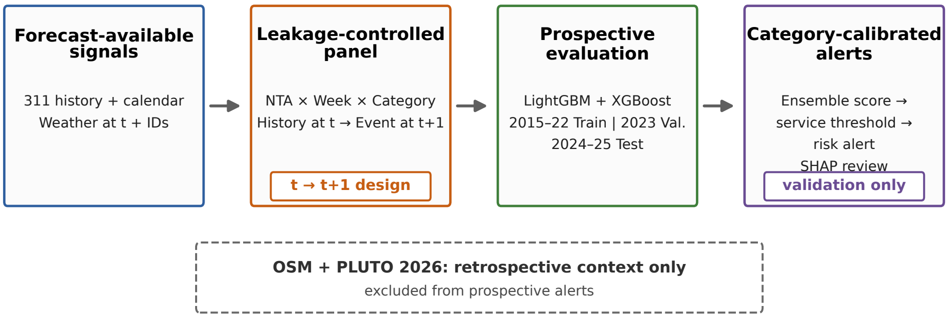

Fig. 1 Overview of the revised early-warning framework. The prospective alerting path uses only predictors available by the end of feature week t, while the target represents abnormal reported demand at $t+1$. Models are evaluated chronologically, and category thresholds are selected on validation. Contemporary OSM and PLUTO snapshots are assessed separately as retrospective context and are not used to produce prospective alerts.

2 Related Work

Urban computing links administrative, geographic, and sensed data to support city analysis and decision making [5]. Recent spatiotemporal learning surveys emphasize the value of temporal structure, spatial context, and prospective validation in urban prediction problems [6]. Within 311 research, prior work shows that complaint structure can distinguish urban locations [1], while reporting bias can distort data-driven governance decisions [2, 3]. Related domain studies connect 311 requests to flooding [7], extreme heat [8], foodborne illness signals [9], open-data curation [4], and municipal service-level design [10]. Our task differs by forecasting future abnormal reported demand across multiple service categories in a single citywide panel.

Contextual urban features can improve service-demand models when interpreted carefully. OSM provides volunteered geographic information and POI records useful for describing neighborhood function, but coverage and tagging are uneven [11, 12]. POI and land-use features have been used for urban change, functional-zone, land-use, and traffic-flow modeling [13–16]. Here, contemporary OSM POIs and PLUTO built-environment summaries are used only for retrospective predictive-association checks; they do not feed prospective alerts.

The forecasting target is an imbalanced binary event, making ordinary accuracy uninformative. Imbalance-learning surveys and evaluation work motivate precision, recall, F1, PR-AUC, ROC-AUC, and balanced accuracy as complementary views [17–21]. Gradient-boosted tree methods such as XGBoost and LightGBM are strong tabular baselines for nonlinear interactions [22, 23]. Because operational use requires transparency, we use SHAP and TreeSHAP to explain a tuned LightGBM component while following cautions that explanations of black-box models should not be mistaken for causal guarantees [24–26].

Table 1 Dataset construction summary.

Item	Value	Item	Value
NTA-matched 311	30,940,129	Match rate	99.9777%
Unit	NTA-week-cat.	Panel rows	1,355,850
Valid-label rows	1,334,628	Model-ready rows	1,325,196
Excluded rows	30,654	NTAs	262
Weeks	575	Categories	9
Coverage	2014-12-29-2025-12-29	NOAA days	4,018
OSM POIs	77,083	Output columns	252
Prospective inputs	61	Prospective columns	73
Context-check inputs	213	Context resources	OSM/PLUTO 2026

3 Methodology

3.1 Data and Task Formulation

The project integrates six public data sources: NYC Open Data 311 Service Requests from the historical 2010–2019 archive and the current 2020–present archive [27, 28], 2020 NTA boundaries [29], NOAA GHCN-Daily weather for Central Park station `GHCND:USW00094728` [30], OSM POIs queried through Overpass [31], and NYC PLUTO/MapPLUTO land-use records [32]. The analysis window uses 2015–2025 records from the combined 311 archives. Weekly 311 counts use Monday week starts and are spatially joined to 2020 NTAs. Weather covariates summarize the feature week t, not the target week $t+1$. OSM and PLUTO variables are fixed extraction snapshots from 2026-06-25; they are excluded from the prospective model and used only in the retrospective context analysis.

The analytical unit is (n, t, c), where n is an NTA, t is a feature-week start date, and c is one of nine stable complaint categories: noise, housing, water/sewer, public safety, parking/traffic, environment, infrastructure, sanitation, and other. These categories reduce sparsity while preserving service meaning. The dense panel contains 262 NTAs, 575 weekly periods, and nine categories, yielding 1,355,850 rows. Relative to the full dense panel, 30,654 rows are excluded due to warm-up/history requirements and unavailable next-week targets, leaving 1,325,196 model-ready rows. Table 1 summarizes the construction.

For each feature week t, all predictors are available after that week closes. The target week is $t+1$, seven days after the feature-week start date:

$$y_{n,t,c} = \mathbf{1}\{\text{count}_{n,t+1,c} > \mu^{8w}_{n,t,c} + 1.5\sigma^{8w}_{n,t,c}\}.$$

The rolling mean $\mu^{8w}_{n,t,c}$ and standard deviation $\sigma^{8w}_{n,t,c}$ are computed from a shifted historical eight-week window. The label therefore identifies next-week abnormal reported demand relative to each local recent baseline.

3.2 Predictive Models and Category-Aware Decision Layer

The final prospective configuration uses 61 inputs: complaint category, borough, current `complaint_count`, lag and rolling features, ratios to shifted historical baselines, history-availability flags, calendar variables, and feature-week weather. One-hot

Table 2 Compact tuning and final-score protocol for gradient-boosted models.

Item	Setting
Feature set	61 prospective inputs; one-hot expansion to 73 model columns; excludes OSM/PLUTO 2026
Tuning sample/seed	Stratified train sample, `max_train_rows` = 300,000, seed 42; selected candidates rebuilt on all 954,990 training rows
LightGBM grid	Depth/leaf/lr: (8,63,0.05), (6,31,0.03), (10,127,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
XGBoost grid	Depth/lr: (8,0.05), (6,0.03), (5,0.03); class weights none, square-root imbalance, half imbalance, and full balanced weight
Selection	Validation F1; selected LightGBM uses 800 estimators, lr 0.03, max depth 6, 31 leaves, and balanced positive weight 6.8726; selected XGBoost uses 900 estimators, lr 0.03, max depth 5, and square-root positive weight 2.6216
Selected settings	LightGBM: binary objective, min_child_samples = 100, subsample = 0.85, colsample_bytree = 0.85, reg_lambda = 3.0. XGBoost: binary:logistic objective, eval_metric = aucpr, min_child_weight = 80, subsample = 0.9, colsample_bytree = 0.9, reg_lambda = 5.0, tree_method = hist. Seed 42 is used for sampling and model fitting
Thresholds	Validation grid 0.01–0.99 step 0.005 plus score quantiles; category fallback to global threshold if validation positives < 30

encoding expands these to 73 model columns. OSM and PLUTO 2026 features are stored in separate retrospective configurations but are not eligible for final model selection, final ensemble scores, category thresholds, headline metrics, or alert production.

We compare rule baselines, logistic regression, LightGBM, XGBoost, and an equal-weight LightGBM/XGBoost ensemble. Numeric medians and categorical levels are fitted on the training period only. Hyperparameters are selected from stratified 300,000-row tuning runs for reproducibility, then final scores are rebuilt on all 954,990 training rows. Table 2 records the compact protocol.

The final settings are read from the LightGBM, XGBoost, and ensemble run summaries under `data/processed/model_results/`. Parameters not explicitly passed by the scripts, including XGBoost `gamma` and `reg_alpha`, retain their library defaults.

Let $s^L_{n,t,c}$ be the tuned LightGBM score and $s^X_{n,t,c}$ the tuned XGBoost score. The ensemble score is $s^E_{n,t,c} = 0.5s^L_{n,t,c} + 0.5s^X_{n,t,c}$. The category-aware decision layer assigns each category c a validation-selected threshold τ_c, producing $\hat{y}_{n,t,c} = \mathbf{1}\{s^E_{n,t,c} \geq \tau_c\}$. This is category-aware at the decision stage: the same probability score can map to different alert decisions because service domains differ in score distribution, prevalence, and operating cost.

3.3 Evaluation and Explainability Protocol

Evaluation uses a chronological split by target week: 2015-2022 for training, 2023 for validation, and 2024-2025 for testing. The model-ready split contains 954,990 train rows with 121,306 positives, 122,616 validation rows with 15,574 positives, and 247,590 test rows with 31,570 positives. Validation and test positive shares are approximately

Table 3 Prospective feature ablation and retrospective context extension under the same compact LightGBM protocol. OSM and PLUTO rows are retrospective context checks using contemporary 2026 snapshots; they are not used in the final prospective model. Thresholds are selected on validation; PR-AUC and ROC-AUC use raw scores.

Feature configuration	Val PR-AUC	Test F1	Prec.	Rec.	Test PR-AUC	ROC-AUC
A. Prospective predictors						
Core history + identifiers	0.2702	0.3513	0.2595	0.5434	0.2838	0.7374
Core + calendar	0.3072	0.3785	0.2873	0.5546	0.3262	0.7635
Core + calendar + feature-week weather	0.3143	0.3767	0.2843	0.5581	0.3246	0.7615
B. Retrospective context extension						
Prospective + OSM 2026	0.3189	0.3818	0.2915	0.5535	0.3314	0.7673
Prospective + PLUTO 2026	0.3165	0.3813	0.3008	0.5208	0.3299	0.7668
Prospective + OSM 2026 + PLUTO 2026	0.3180	0.3819	0.3066	0.5061	0.3297	0.7679

12.7%. All thresholds are selected on validation only and evaluated unchanged on the held-out test period.

F1 is the primary validation-ranking metric because the alerting task requires balancing event detection and alert conversion. We also report precision, recall, PR-AUC, ROC-AUC, balanced accuracy, alert rate, and confusion matrices. Threshold tuning is treated as a method component because it changes hard-alert metrics and alert volume; it does not change PR-AUC or ROC-AUC for a fixed score vector.

For explainability, SHAP is computed for the tuned prospective LightGBM component. Since the final evaluated score is an ensemble with category-specific thresholds, SHAP values are interpreted as evidence about a representative tree component under the same prospective feature design and chronological protocol, not as a direct explanation of every final alert or as causal attribution.

4 Results

4.1 Overall Performance and Ablation Results

Rule and logistic baselines confirm that simple history alone is not enough for the final alerting task. On the 2024-2025 target-week test period, current-week persistence obtains F1 = 0.3111 and PR-AUC = 0.2359, a seasonal lag-52 margin obtains F1 = 0.2423 and PR-AUC = 0.1773, and the strongest rule baseline, a rolling z-score rule, obtains F1 = 0.3332 and PR-AUC = 0.2658. Logistic regression with historical features obtains F1 = 0.3256 and PR-AUC = 0.2598.

Table 3 reports a controlled compact LightGBM comparison using the same target-week split, validation-selected threshold policy, and 300,000-row stratified training sample for each configuration. Calendar provides the largest prospective gain over core history. The OSM/PLUTO rows are retrospective context checks only: they show small additional predictive association in this compact protocol but are not used for final ensemble scores, thresholds, alerts, or headline metrics.

Table 4 compares final prospective decision strategies. Following the validation-selection rule, the selected strategy is the LightGBM/XGBoost ensemble with complaint-category thresholds. On the held-out future test period, it achieves F1 = 0.3802, precision = 0.2933, recall = 0.5404, PR-AUC = 0.3310, ROC-AUC = 0.7643, and balanced accuracy = 0.6750. The category-specific layer is retained because it exposes service-specific operating points selected only on validation; the result should

Table 4 Final prospective decision-layer comparison. Thresholds are selected on validation and evaluated unchanged on the future test period.

Model	Threshold	Val F1	Test F1	Prec.	Rec.	PR-AUC	ROC-AUC	Bal. acc.	Alert rate
Ensemble LGBM(0.50)+XGB(0.50)	Per category	0.3764	0.3802	0.2933	0.5404	0.3310	0.7643	0.6750	0.2349
Tuned XGBoost	Per category	0.3762	0.3802	0.2973	0.5270	0.3306	0.7642	0.6725	0.2260
Tuned LightGBM	Per category	0.3740	0.3793	0.2940	0.5343	0.3298	0.7636	0.6734	0.2317
Ensemble LGBM(0.50)+XGB(0.50)	Global	0.3733	0.3801	0.2967	0.5287	0.3310	0.7643	0.6728	0.2272

Table 5 Final prospective model test performance by complaint category on the 2024-2025 target-week period.

Category	Thr.	Positives	Alerts	PR-AUC	F1	Precision	Recall
Noise	0.5100	3,982	6,205	0.4159	0.4555	0.3739	0.5826
Public safety	0.4160	3,536	7,391	0.3269	0.4058	0.3000	0.6270
Housing	0.4950	3,720	5,268	0.3897	0.4010	0.3421	0.4844
Water/sewer	0.4750	3,462	6,516	0.3542	0.3852	0.2950	0.5552
Parking/traffic	0.4385	3,586	7,735	0.3208	0.3696	0.2705	0.5834
Other	0.4512	3,620	6,691	0.3104	0.3668	0.2826	0.5224
Environment	0.4600	2,896	5,562	0.2991	0.3556	0.2704	0.5193
Infrastructure	0.4400	3,596	7,692	0.2710	0.3540	0.2598	0.5556
Sanitation	0.4400	3,172	5,105	0.2525	0.3173	0.2572	0.4139

be interpreted as the validation-selected category-specific strategy, not as evidence of aggregate metric dominance.

The final prospective test confusion matrix contains 17,059 true positives, 41,106 false positives, 14,511 false negatives, and 174,914 true negatives. The top 1% of scored test rows has precision 0.5905 and 4.63-times lift over the test base rate; the top 5% has precision 0.4460; and the top 10% has precision 0.3809. These concentration results position the model as prioritization support rather than an automatic allocation rule.

4.2 Category-Level Alert Analysis

Performance varies across municipal service domains (Table 5). Noise has the strongest F1, followed by public safety, housing, and water/sewer. Positives are observed abnormal next-week rows within the held-out test set; alerts are rows whose prospective ensemble score exceeds the validation-selected threshold for that complaint category. Summing the category positives gives 31,570 and summing the category alerts gives 58,165, matching the overall confusion matrix.

Figure 2 makes the operating-point trade-off visible. Noise has more observed positives than public safety but fewer alerts; its higher threshold yields higher precision and slightly lower recall, while public safety issues more alerts to capture higher recall at lower precision. This is a consequence of category thresholds and score distributions. The result should be read as the pre-specified category-specific strategy selected on validation, not as the best aggregate-F1 strategy.

Target-definition sensitivity was checked with a compact LightGBM prospective protocol using no OSM/PLUTO features, the same chronological split, and validation-selected thresholds (Table 6). Event prevalence changes as expected: a 1.0-standard-deviation threshold marks more rows positive, while 2.0 standard deviations yields a rarer target. These compact runs support within-table comparison only and are not directly comparable with the final full-training ensemble.

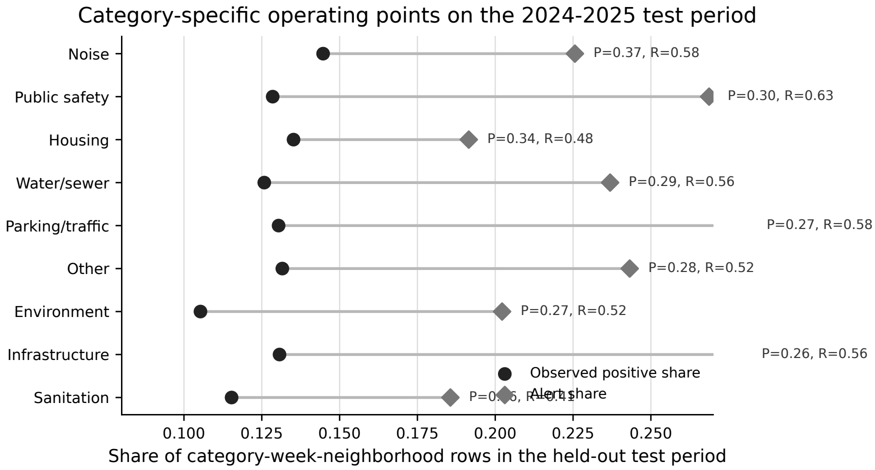

Fig. 2 Observed positive share and validation-threshold alert share by complaint category in the held-out test period. Labels show category precision and recall at the selected operating point.

Table 6 Sensitivity to abnormal-demand definition under a compact LightGBM prospective protocol. Scores are for within-table comparison and are not directly comparable with the final full-training ensemble in Table 4.

Target definition	Pos. share	F1	PR-AUC	ROC-AUC
4w + 1.5σ	0.1443	0.4144	0.3640	0.7689
8w + 1.0σ	0.1833	0.4419	0.3986	0.7485
8w + 1.5σ (reference)	0.1275	0.3741	0.3196	0.7572
8w + 2.0σ	0.0885	0.3170	0.2527	0.7655
12w + 1.5σ	0.1180	0.3646	0.3155	0.7642

4.3 Explanation Results

Figure 3 summarizes SHAP importance for the prospective LightGBM component. Historical temporal features dominate, accounting for 75.15% of summed mean absolute SHAP importance across 21 features. Service category accounts for 8.16%, calendar for 6.13%, weather for 5.87%, current demand for 2.74%, and spatial identifiers for 1.23%. The remaining 0.72% is assigned to other low-contribution predictors. OSM and PLUTO do not appear because they are not final prospective inputs.

Figure 4 shows the leading individual features. The strongest terms are historical: `rolling_8w_std`, `ratio_to_8w_mean`, `lag_52w_count`, `rolling_12w_mean`, `rolling_12w_std`, `rolling_4w_mean`, and `lag_1w_count`. Calendar timing appears through `week_of_year`, current demand appears through `complaint_count`, and complaint-category indicators enter the top ranks. SHAP explains the LightGBM component, not every ensemble alert, and should be read as predictive association rather than causal effect.

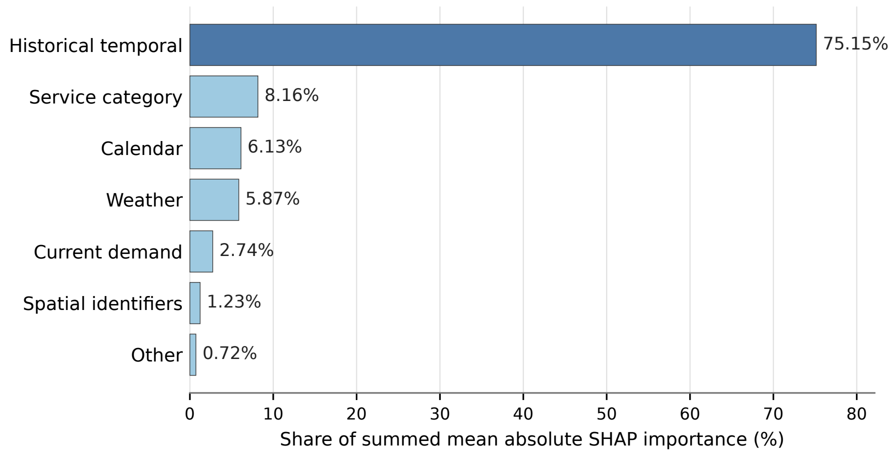

Fig. 3 Share of summed mean absolute SHAP importance by feature group for the prospective LightGBM component. Historical temporal features dominate, while service category, calendar, weather, current demand, and spatial identifiers provide smaller predictive contributions.

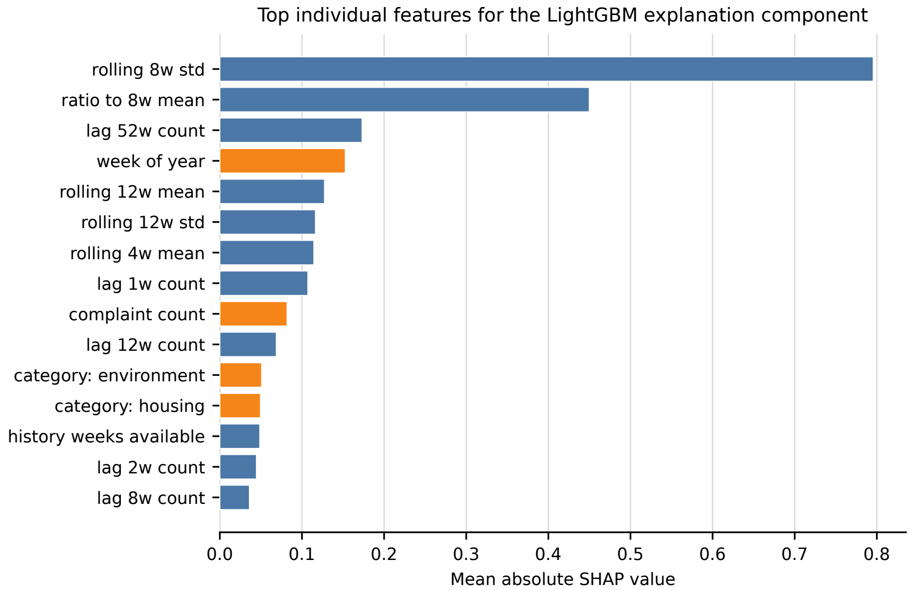

Fig. 4 Top individual SHAP features for the prospective LightGBM explanation component. Blue bars are historical temporal features; orange bars are other feature groups.

Local SHAP cases are recomputed from the prospective model and support analyst review of alerts and errors using only historical, calendar, weather, service, and geographic identifiers.

5 Discussion

5.1 Operational and Methodological Implications

The model is most useful as early-warning decision support. Recall of 0.5404 means it captures slightly more than half of future abnormal reported-demand events, while precision of 0.2933 means many alerts are false positives. Agencies should therefore use alerts to prioritize review, pre-position attention, or monitor risk queues, with final actions incorporating severity, staffing, and local knowledge.

Category thresholds make the alert layer more transparent because service domains do not share the same base rates, volatility, response costs, or tolerance for missed events. The final alerts do not use contemporary OSM/PLUTO snapshots; they rely on demand history, calendar timing, feature-week weather, service category, and borough identifiers. The retrospective context analysis only indicates additional association from 2026 context descriptors and should not be turned into an operational deployment claim. Sensitivity checks show that target prevalence changes under alternative abnormal-demand definitions, so thresholds and workload should be revisited if an agency adopts a different event definition.

5.2 Limitations and Future Work

Several limits should guide interpretation. NYC 311 data measure reported service demand, not all underlying urban problems, and reporting propensity can reproduce socio-spatial inequities. Contemporary OSM and PLUTO snapshots are excluded from prospective alerts; their retrospective analysis cannot establish historical or causal context effects. Time-matched historical snapshots are still needed to evaluate urban context prospectively. NOAA weather is weekly city-level exposure rather than NTA microclimate. Hyperparameters are selected from sampled tuning runs, category thresholds optimize validation F1 rather than agency-specific costs, and full operational utility would require response-time, staffing, service-outcome, and fairness data. Future work should test historical OSM/PLUTO, spatial weather, calibrated probabilities, cost-sensitive thresholds, alternative targets, cross-city transfer, and operational outcome evaluation.

6 Conclusion

An explainable forecasting framework was developed for next-week abnormal reported demand in NYC 311 data at the neighborhood, week, and service-category level. The final prospective model uses only forecast-available predictors: shifted demand history, calendar timing, feature-week weather, service category, and borough identifiers. On a strict 2024-2025 future test, the validation-selected LightGBM/XGBoost ensemble reaches F1 = 0.3802, precision = 0.2933, recall = 0.5404, and PR-AUC = 0.3310. SHAP shows that historical dynamics dominate, while category, calendar, and weather features refine risk estimates. Category thresholds make service-domain alert trade-offs explicit. OSM/PLUTO context is retained only as a separate retrospective extension, and the main limitation remains that 311 data forecast reported demand rather than all urban need.

Declarations

Data and code availability. Source data are NYC Open Data 311 historical archive `76ig-c548`, NYC Open Data 311 current archive `erm2-nwe9`, NOAA GHCN-Daily station `GHCND:USW00094728`, OpenStreetMap Overpass, NYC PLUTO/MapPLUTO `64uk-42ks`, and 2020 NTA boundaries `9nt8-h7nd`; the local snapshot was extracted on 2026-06-25. Scripts, mappings, configurations, and evaluation code are available at `https://github.com/PhongLam26/SIMC_NYC`; processed data release is subject to licensing, redistribution, and storage constraints.

Competing interests. The authors declare no competing interests.

Funding. No specific funding was received.

Author contributions. Tran Dai Phong Lam led data processing, modeling, analysis, and drafting; Thu Le served as academic advisor and supervised the study design, methodology, and manuscript editing; Nguyen Quoc Hung and Nguyen Trung Trinh contributed validation and review.

References

[1] Wang, L., Qian, C., Kats, P., Kontokosta, C.E., Sobolevsky, S.: Structure of 311 service requests as a signature of urban location. PLOS ONE **12**(10), e0186314 (2017) https://doi.org/10.1371/journal.pone.0186314

[2] Kontokosta, C.E., Hong, B., Korsberg, K.: Equity in 311 Reporting: Understanding Socio-Spatial Differentials in the Propensity to Complain (2017). https://arxiv.org/abs/1710.02452

[3] Kontokosta, C.E., Hong, B.: Bias in smart city governance: How socio-spatial disparities in 311 complaint behavior impact the fairness of data-driven decisions. Sustainable Cities and Society **64**, 102503 (2021) https://doi.org/10.1016/j.scs.2020.102503

[4] Tussey, D., Yan, J.: Principles for Open Data Curation: A Case Study with the New York City 311 Service Request Data (2025). https://arxiv.org/abs/2502.08649

[5] Zheng, Y., Capra, L., Wolfson, O., Yang, H.: Urban computing: Concepts, methodologies, and applications. ACM Transactions on Intelligent Systems and Technology **5**(3), Article 38, 1–55 (2014) https://doi.org/10.1145/2629592

[6] Jin, G., Liang, Y., Fang, Y., Shao, Z., Huang, J., Zhang, J., Zheng, Y.: Spatio-temporal graph neural networks for predictive learning in urban computing: A survey. IEEE Transactions on Knowledge and Data Engineering **36**(10), 5388–5408 (2024) https://doi.org/10.1109/TKDE.2023.3333824

[7] Agonafir, C., Ramirez Pabon, A., Lakhankar, T., Khanbilvardi, R., Devineni, N.: Understanding New York City street flooding through 311 complaints. Journal of Hydrology **605**, 127300 (2022) https://doi.org/10.1016/j.jhydrol.2021.127300

[8] Kianmehr, A., Pamukcu, D.: Analyzing citizens' needs during an extreme heat event, based on 311 service requests: A case study of the 2021 heatwave in vancouver, british columbia. In: ISCRAM 2022 Conference Proceedings – 19th International Conference on Information Systems for Crisis Response and Management, pp. 174–182 (2022). https://idl.iscram.org/files/aydakianmehr/2022/2408_AydaKianmehr+DuyguPamukcu2022.pdf

[9] Shaveet, E., Su, C., Hsu, D., Gravano, L.: Seasonality Patterns in 311-Reported Foodborne Illness Cases and Machine Learning-Identified Indications of Foodborne Illnesses from Yelp Reviews, New York City, 2022-2023 (2024). https://arxiv.org/abs/2405.06138

[10] Liu, Z., Garg, N.: Redesigning Service Level Agreements: Equity and Efficiency in City Government Operations (2024). https://arxiv.org/abs/2410.14825

[11] Haklay, M., Weber, P.: OpenStreetMap: User-generated street maps. IEEE Pervasive Computing **7**(4), 12–18 (2008) https://doi.org/10.1109/MPRV.2008.80

[12] Vargas-Munoz, J.E., Srivastava, S., Tuia, D., Falcao, A.X.: OpenStreetMap: Challenges and opportunities in machine learning and remote sensing. IEEE Geoscience and Remote Sensing Magazine **9**(1), 184–199 (2021) https://doi.org/10.1109/MGRS.2020.2994107

[13] Zhang, L., Pfoser, D.: Using OpenStreetMap point-of-interest data to model urban change—a feasibility study. PLOS ONE **14**(2), e0212606 (2019) https://doi.org/10.1371/journal.pone.0212606

[14] Luo, G., Ye, J., Wang, J., Wei, Y.: Urban functional zone classification based on POI data and machine learning. Sustainability **15**(5), 4631 (2023) https://doi.org/10.3390/su15054631

[15] Zhou, Q., Cao, R.: A coarse-to-fine approach for urban land use mapping based on multisource geospatial data. In: 2022 29th International Conference on Geoinformatics (2022). https://doi.org/10.1109/GEOINFORMATICS57846.2022.9963851

[16] Wang, K., Liu, L., Liu, Y., Li, G., Zhou, F., Lin, L.: Urban regional function guided traffic flow prediction. Information Sciences **634**, 308–320 (2023) https://doi.org/10.1016/j.ins.2023.03.109

[17] He, H., Garcia, E.A.: Learning from imbalanced data. IEEE Transactions on Knowledge and Data Engineering **21**(9), 1263–1284 (2009) https://doi.org/10.1109/TKDE.2008.239

[18] Fawcett, T.: An introduction to ROC analysis. Pattern Recognition Letters **27**(8), 861–874 (2006) https://doi.org/10.1016/j.patrec.2005.10.010

[19] Davis, J., Goadrich, M.: The relationship between precision-recall and ROC curves. In: Proceedings of the 23rd International Conference on Machine Learning, pp. 233–240 (2006). https://doi.org/10.1145/1143844.1143874

[20] Powers, D.M.W.: Evaluation: From precision, recall and f-measure to ROC, informedness, markedness and correlation. Journal of Machine Learning Technologies **2**(1), 37–63 (2011)

[21] Saito, T., Rehmsmeier, M.: The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. PLOS ONE **10**(3), e0118432 (2015) https://doi.org/10.1371/journal.pone.0118432

[22] Chen, T., Guestrin, C.: XGBoost: A scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. ACM, New York, NY, USA (2016). https://doi.org/10.1145/2939672.2939785

[23] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., Liu, T.-Y.: LightGBM: A highly efficient gradient boosting decision tree. In: Advances in Neural Information Processing Systems, vol. 30 (2017). https://papers.nips.cc/paper/6907-lightgbm-a-highly-efficient-gradient-boosting-decision-tree

[24] Lundberg, S.M., Lee, S.-I.: A unified approach to interpreting model predictions. In: Advances in Neural Information Processing Systems, vol. 30 (2017). https://arxiv.org/abs/1705.07874

[25] Lundberg, S.M., Erion, G., Chen, H., DeGrave, A., Prutkin, J.M., Nair, B., Katz, R., Himmelfarb, J., Bansal, N., Lee, S.-I.: From local explanations to global understanding with explainable AI for trees. Nature Machine Intelligence **2**, 56–67 (2020) https://doi.org/10.1038/s42256-019-0138-9

[26] Rudin, C.: Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. Nature Machine Intelligence **1**, 206–215 (2019) https://doi.org/10.1038/s42256-019-0048-x

[27] City of New York: 311 Service Requests from 2010 to 2019. NYC Open Data (2026). https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2010-to-2019/76ig-c548

[28] City of New York: 311 Service Requests from 2020 to Present. NYC Open Data (2026). https://data.cityofnewyork.us/Social-Services/311-Service-Requests-from-2020-to-Present/erm2-nwe9

[29] New York City Department of City Planning: 2020 Neighborhood Tabulation Areas (NTAs). NYC Open Data (2026). https://data.cityofnewyork.us/City-Government/2020-Neighborhood-Tabulation-Areas-NTAs-/9nt8-h7nd

[30] Menne, M.J., Durre, I., Vose, R.S., Gleason, B.E., Houston, T.G.: An overview of the global historical climatology network-daily database. Journal of Atmospheric and Oceanic Technology **29**(7), 897–910 (2012) https://doi.org/10.1175/JTECH-D-11-00103.1

[31] OpenStreetMap Wiki Contributors: Overpass API (2026). https://wiki.openstreetmap.org/wiki/Overpass_API

[32] New York City Department of City Planning: Primary Land Use Tax Lot Output (PLUTO). NYC Open Data (2026). https://data.cityofnewyork.us/City-Government/Primary-Land-Use-Tax-Lot-Output-PLUTO-/64uk-42ks